

2027 SEAK Pink Salmon Forecast

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Objective

To forecast the Southeast Alaska (SEAK) pink salmon commercial harvest in 2027. This document is for guidance as to what was done for the current forecast year. It is for internal use only.

Executive Summary

Forecasts were developed using an approach originally described in Wertheimer et al. (2006), and modified in Orsi et al. (2016) and Murphy et al. (2019), but assuming a log-normal error structure (Miller et al. 2022). This approach is based on a multiple linear regression model with either; 1) the raw juvenile pink salmon CPUE (adj_raw_pink_log; a proxy for abundance) or 2) the vessel-adjusted juvenile pink salmon CPUE (adjusted by the pooled-species vessel calibration coefficient for the Cobb; vessel-adjusted CPUE; a proxy for abundance). Additional variables included a vessel factor to account for the survey vessels through time, an odd and even year factor to account for potential odd and even year cycles of abundance, and temperature data from either the Southeast Alaska Coastal Monitoring Survey (SECM; Piston et al. 2021; ISTI20_JJ) or from satellite-derived sea surface temperature (SST) data (Huang et al. 2017).

There were 36 individual models considered:

- adj_raw_pink_log model with a vessel interaction, an odd/even year factor, and a vessel factor (m1);
- adj_raw_pink_log model with a vessel interaction, an odd/even year factor, a vessel factor, and temperature data from the the SECM survey (m2);
- adj_raw_pink_log models with a vessel interaction, an odd/even year factor, a vessel factor, and satellite SST data (m3-m18);
- vessel-adjusted CPUE, and an odd/even year factor (m1a);

- vessel-adjusted CPUE, an odd/even year factor, and temperature data from the SECM survey (m2a); and
- vessel-adjusted CPUE models with an odd/even year factor, and satellite-derived SST data (m3a-m18a).

The `adj_raw_pink_log` variable is defined as the natural logarithm of the maximum juvenile pink salmon catch, adjusted to a 20 minute haul, from either June or July. This CPUE variable is not vessel-calibrated. The Stellar and Chellissa vessels were only used for one year each, 2008 and 2009, respectively, and so those two years are not used in the assessment models that incorporated an `adj_raw_pink_log` variable (Table 1).

The vessel-adjusted CPUE variable is defined as the average natural log transformed CPUE (adjusted to a 20 minute haul) for juvenile pink salmon catches in either June or July, whichever month had the highest catch in a given year, adjusted by the pooled-species vessel calibration coefficient from the R/V Cobb. As there was not a SECM survey in May, temperature data from May was unavailable. Therefore, the temperature variable from the SECM survey (`ISTI20_JJ`) was the average from June and July. In prior forecasting years, the `ISTI` variable was the average from May through July.

The model performance metric one-step ahead mean absolute percent error (MAPE) for the last five years (MAPE5; forecast years 2021 through 2025) was used to evaluate the forecast accuracy of the 36 individual models, the AICc values were calculated for each model to prevent over-parameterization of the model, and the adjusted R-squared values, significant terms, and overall p-value of the model were used to determine fit. As the model data sets differ between the models m1 through m18 and the models m1a through m18a, AICc values can not be compared across these models. The model performance metric MAPE is a standard statistical metric used to measure the accuracy of forecasting and prediction models. It calculates the average of the absolute percentage differences between predicted values and actual outcomes. Because it measures error as a relative percentage rather than a raw number, a lower MAPE indicates higher accuracy, while a higher MAPE indicates a less reliable model. The MAPE value can be compared across the 36 models. Based upon the performance metric the 5-year MAPE, the AICc values, significant parameters in the models, and the adjusted R-squared values, model mxx (a model that included xxx; Appendix C) was the best performing model and the 2027-forecast using this model has a point estimate of xx3 million fish (80% prediction interval: xx to xx million fish).

Analysis

Individual, multiple linear regression models

Biophysical variables based on data from Southeast Alaska (Table 1) were used to forecast the harvest of adult pink salmon in Southeast Alaska, one year in advance, using individual,

multiple linear regression models with two main structures (models m1-m18 and models m1a-m18a). The first model structure, model m1, consisted of the predictor variable juvenile adj_raw_pink_log (CPUE) with a vessel factor interaction, an odd/even year factor, and a vessel factor (model m1),

$$E(Y) = \beta_0 + \beta_1 X_{V1} + \beta_2 X_{V2} + \beta_3 X_O + \beta_4 CPUE + \beta_5 (X_{V1} \times CPUE) + \beta_6 (X_{V2} \times CPUE) + \epsilon.$$

The reference category is an even year with the R/V Cobb. The vessel factor and even/odd year factor adjust the intercept, and the vessel interaction terms adjust the model slope. For example, during even years when the research vessel is the Cobb, the model simplifies to $E(Y) = \beta_0 + \beta_4 CPUE + \epsilon$. Although the simplest model does not contain a temperature variable, including temperature data with CPUE has been shown to result in a substantial improvement in the accuracy of model predictions (Murphy et al. 2019). The temperature index for models m2-m18 was either the SECM survey Icy Strait temperature Index (ISTI20_JJ; Murphy et al. 2019) or one of the 16 satellite-derived SST data (Huang et al. 2017; definitions in Appendix A).

The second model structure, model m1a, consisted of the predictor variable vessel-adjusted CPUE (CPUE), and an odd/even year factor,

$$E(Y) = \beta_0 + \beta_2 X_O + \beta_3 CPUE + \epsilon.$$

The reference category (i.e., reference intercept) is an even year. The temperature index for models m2a-m18a was either the SECM survey Icy Strait temperature Index (ISTI20_JJ; Murphy et al. 2019) or one of the 16 satellite-derived SST data (Huang et al. 2017).

The response variable (Y ; Southeast Alaska adult pink salmon harvest in millions), the adj_raw_pink_log, and the vessel-adjusted CPUE were log transformed in the model, but temperature data were not. The forecast ($\hat{Y}_i$), and 80% prediction intervals (based on output from program R; R Core Team 2025) from the 36 regression models were exponentiated and bias-corrected (Miller 1984),

$$\hat{F}_i = \exp(\hat{Y}_i + \frac{\sigma_i^2}{2}), \quad (2)$$

where $\hat{F}_i$ is the preseason forecast (for each model i) in millions of fish, and σ_i is the variance (for each model i).

Performance metric: One-step ahead MAPE

The model summary results using the performance metric one-step ahead MAPE are shown in Table 2 and Table 3; the smallest value is the preferred model. The performance metric one-step ahead MAPE was calculated as follows.

1. Estimate the regression parameters at time $t-1$ from data up to time $t-1$.
2. Make a prediction of $\hat{Y}_t$ at time t based on the predictor variables at time t and the estimate of the regression parameters at time $t-1$ (i.e., the fitted regression equation).
3. Calculate the MAPE based on the prediction of $\hat{Y}_t$ at time t and the observed value of Y_t at time t ,

$$\text{MAPE} = \left| \frac{\exp(Y_t) - \exp(\hat{Y}_t + \frac{\sigma_t^2}{2})}{\exp(Y_t)} \right|. \quad (3)$$

4. For each individual model, average the MAPEs calculated from the forecasts,

$$\frac{1}{n} \sum_{t=1}^n \left| \frac{\exp(Y_t) - \exp(\hat{Y}_t + \frac{\sigma_t^2}{2})}{\exp(Y_t)} \right|, \quad (4)$$

where n is the number of forecasts in the average (5 forecasts for the 5-year MAPE and 10 forecasts for the 10-year MAPE). For example, to calculate the five year one-step-ahead MAPE for model m1 for the 2022 forecast, use data up through year 2016 (e.g., data up through year 2016 is $t-1$ and the forecast is for t , or year 2017). Then, calculate a MAPE based on the 2017 forecast and the observed pink salmon harvest in 2017 using equation 3. Next, use data up through year 2017 (e.g., data up through year 2017 is $t-1$ and the forecast is for year 2018; t) and calculate a MAPE based on the 2018 forecast and the observed pink salmon harvest in 2018 using equation 3. Repeat this process for each subsequent year through year 2020 to forecast 2021. Finally, average the five MAPEs to calculate a five year one-step-ahead MAPE for model m1. As the results of the 5-year MAPEs with or without the forecast bias adjustment have been similar (i.e., the model performance order did not change whether the five year one-step-ahead MAPE or the bias-corrected five year one-step-ahead MAPE was compared), for simplicity, the bias adjustment for the forecast was not used in the calculation of the five year one-step-ahead MAPE for model comparison.

Akaike Information Criterion corrected for small sample sizes (AICc)

Hierarchical models were compared with the AICc criterion. The best fit models, according to the AICc criterion, is one that explains the greatest amount of variation with the fewest independent variables (i.e., the most parsimonious; Table 2 and Table 3). Lower AICc values are

preferred, and the AICc criterion penalizes models that use more parameters. Comparing the AICc values of two hierarchical models, a $\Delta_i \leq 2$ suggests that the two models are essentially the same, and the most parsimonious model should be chosen (Burnham and Anderson 2004). If the $\Delta_i > 2$, the model with the lower AICc value should be chosen. As the data set of the response variable is reduced for the model set m1-m18 compared to the model set m1a-m18a, AICc can only be used to compare within the nested models (e.g., among m1-m18).

Adjusted R-squared

The adjusted R-squared is a modified version of the R-squared value that adjusts for the number of predictors in the model and prevents overfitting by balancing model performance and complexity. A higher adjusted R-squared value is preferred. As the adjusted R-squared metric depends on both sample size and the variance of the dependent variable in the data set, the adjusted R-squared can only be compared for the models with the same response variable data set. As the data set for the model set m1-m18 is reduced, the adjusted R-squared can only be compared within the model group (i.e., among m1-m18).

Results

Based upon the 5-year MAPE performance metric, the AICc values, significant parameters in the models, and the adjusted R-squared values, model mxx (a model that included xxxx Table 5; Appendix C) was the best performing model. The 2027-forecast using this model has a point estimate of xx million fish (80% prediction interval: xx to xx million fish).

Table 1: Summary of the data used for the 2027-forecast. The Stellar and Chellissa vessels were only used for one year each, 2008 and 2009, respectively, and so those two years are not used in the assessment models that incorporated an `adj_raw_pink_log` variable. The CPUE variable is the vessel-adjusted CPUE, JYear is the juvenile year, and Year is the calendar year.

JYear	Year	Harvest	odd_even_factor	vessel	adj_raw_pink_log	CPUE
1997	1998	42.4	even	Cobb	5.33	2.48
1998	1999	77.8	odd	Cobb	7.14	5.62
1999	2000	20.2	even	Cobb	5.05	1.60
2000	2001	67.0	odd	Cobb	6.39	3.73
2001	2002	45.3	even	Cobb	5.50	2.87
2002	2003	52.5	odd	Cobb	6.11	2.78
2003	2004	45.3	even	Cobb	5.32	3.08
2004	2005	59.1	odd	Cobb	6.80	3.90
2005	2006	11.6	even	Cobb	5.06	2.04

JYear	Year	Harvest	odd_even_factor	vessel	adj_raw_pink_log	CPUE
2006	2007	44.8	odd	Cobb	6.14	2.57
2007	2008	15.9	even	Cobb	3.85	1.17
2008	2009	38.0	odd	Steller	6.64	2.32
2009	2010	24.1	even	Chellissa	6.19	2.33
2010	2011	58.9	odd	NW Explorer	9.43	4.11
2011	2012	21.3	even	NW Explorer	5.71	1.45
2012	2013	94.7	odd	NW Explorer	8.52	3.52
2013	2014	37.2	even	NW Explorer	6.80	2.14
2014	2015	35.1	odd	NW Explorer	7.67	3.82
2015	2016	18.4	even	NW Explorer	6.72	2.45
2016	2017	34.7	odd	NW Explorer	7.94	4.35
2017	2018	8.1	even	NW Explorer	3.56	0.35
2018	2019	21.1	odd	Medeia	3.93	1.17
2019	2020	8.1	even	Medeia	5.32	1.14
2020	2021	48.5	odd	Medeia	4.41	2.15
2021	2022	18.3	even	Medeia	4.46	0.87
2022	2023	48.5	odd	Medeia	4.33	1.45
2023	2024	20.1	even	Medeia	3.40	1.19
2024	2025	21.4	odd	Medeia	3.85	1.66
2025	2026	20.4	even	Medeia	5.02	1.29
2026	2027	NA	odd	Medeia	6.14	3.51

Table 2: Summary of the performance metrics for the 18 regression models with adj_raw_pink_log. The AICc value is the change in AICc value.

Terms	Model	Fit	Fit_LPI	Fit_UPI	AdjR2	MAPE5	AICc_change
Icy_Strait_SST_May	m7	29.1	12.4	68.0	0.67	43.5	1.2
NSEAK_SST_May	m11	28.4	12.3	65.7	0.68	46.0	0.7
ISTI20_JJ	m2	23.2	9.9	54.4	0.65	46.1	2.8
Chatham_SST_May	m3	27.7	12.0	63.9	0.68	46.4	0.8
NSEAK_SST_AMJ	m13	30.0	13.0	68.9	0.69	46.6	0.0
Icy_Strait_SST_AMJ	m9	32.2	13.7	75.8	0.68	47.2	0.6
NSEAK_SST_AMJJ	m14	30.1	12.7	71.5	0.67	48.2	1.6
Chatham_SST_AMJ	m5	28.5	12.5	65.1	0.69	48.4	0.0
SEAK_SST_May	m15	24.6	10.7	56.4	0.67	49.3	1.2
Icy_Strait_SST_AMJJ	m10	31.6	13.0	76.4	0.66	49.5	2.1
Chatham_SST_AMJJ	m6	28.4	12.1	67.0	0.67	49.9	1.7
SEAK_SST_AMJ	m17	26.1	11.4	59.3	0.68	50.3	0.4
SEAK_SST_AMJJ	m18	26.6	11.4	61.8	0.67	50.3	1.5
NSEAK_SST_MJJ	m12	27.6	11.4	66.8	0.65	52.2	3.2
Icy_Strait_SST_MJJ	m8	28.6	11.6	70.2	0.64	53.1	3.6
SEAK_SST_MJJ	m16	24.5	10.4	58.0	0.65	53.1	3.0
Chatham_SST_MJJ	m4	26.3	10.9	63.6	0.64	53.5	3.7
no temperature index included	m1	19.6	7.9	48.5	0.60	54.4	3.5

Table 3: Summary of the performance metrics for the 18 regression models with vessel-adjusted CPUE. The AICc value is the change in AICc value.

Terms	Model	Fit	Fit_LPI	Fit_UPI	AdjR2	MAPE5	AICc_change
ISTI20_JJ	m2a	67.5	41.1	110.8	0.69	21.0	9.8
Chatham_SST_AMJ	m5a	69.5	45.6	106.1	0.77	22.3	0.8
Icy_Strait_SST_AMJ	m9a	80.4	51.9	124.8	0.77	22.7	1.6
NSEAK_SST_AMJ	m13a	66.7	43.5	102.3	0.77	24.1	1.6
Chatham_SST_May	m3a	74.7	49.0	113.7	0.78	24.8	0.0
NSEAK_SST_May	m11a	69.2	45.5	105.3	0.77	25.2	0.5
Icy_Strait_SST_AMJJ	m10a	79.5	49.4	128.1	0.73	25.3	6.0
Icy_Strait_SST_May	m7a	80.6	52.3	124.3	0.77	25.7	0.9
SEAK_SST_May	m15a	60.6	39.4	93.1	0.76	26.2	2.4
SEAK_SST_AMJ	m17a	57.2	37.0	88.5	0.75	26.5	3.4
Chatham_SST_AMJJ	m6a	70.1	44.2	111.1	0.73	27.6	5.6
no temperature index included	m1a	56.8	33.2	97.3	0.62	28.1	13.9
NSEAK_SST_AMJJ	m14a	67.7	42.6	107.5	0.73	28.8	6.1
Icy_Strait_SST_MJJ	m8a	75.4	45.7	124.3	0.70	29.1	9.0
Chatham_SST_MJJ	m4a	69.3	42.4	113.4	0.70	30.5	9.2
SEAK_SST_AMJJ	m18a	59.0	37.1	93.8	0.72	30.7	6.9
NSEAK_SST_MJJ	m12a	65.5	40.4	106.4	0.70	31.1	8.8
SEAK_SST_MJJ	m16a	58.1	35.9	94.1	0.70	33.0	9.3

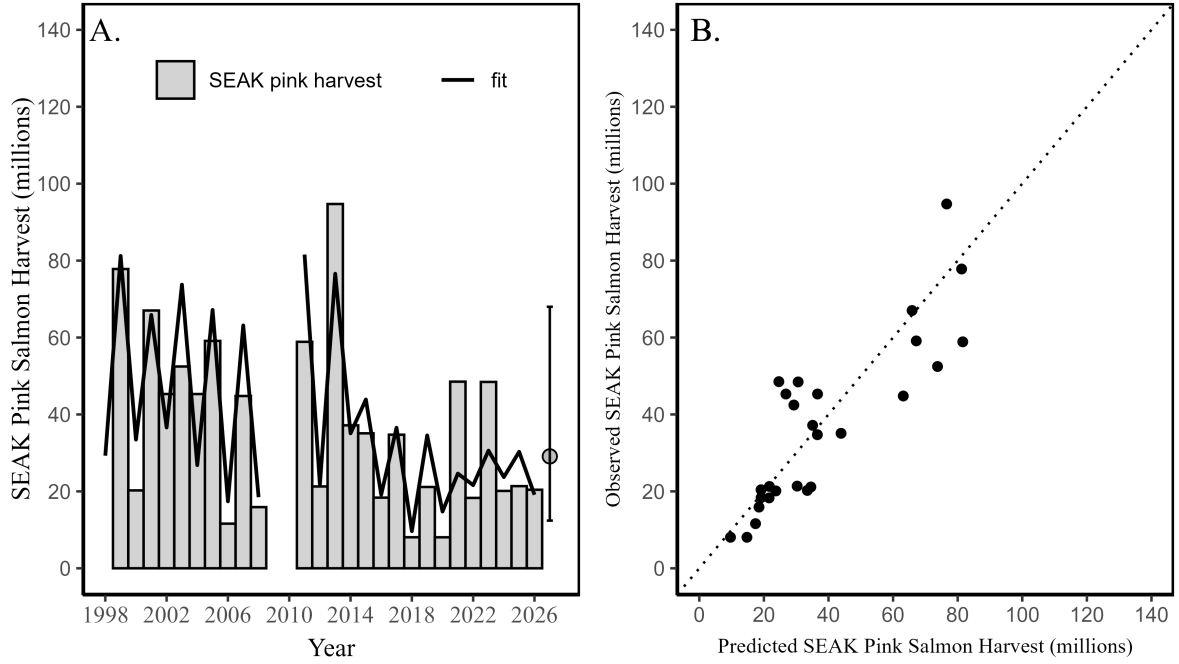


Figure 1: A. SEAK pink salmon harvest (millions) by year with the model fit (line) based upon the best performing model (model m7). The predicted 2027 forecast (29.1 million fish) is symbolized as a grey circle with an 80% prediction interval (12.4 to 68.0 million fish). B. SEAK pink salmon harvest (millions) against the fitted values from model m7 by year. The dotted line is a one to one reference line.

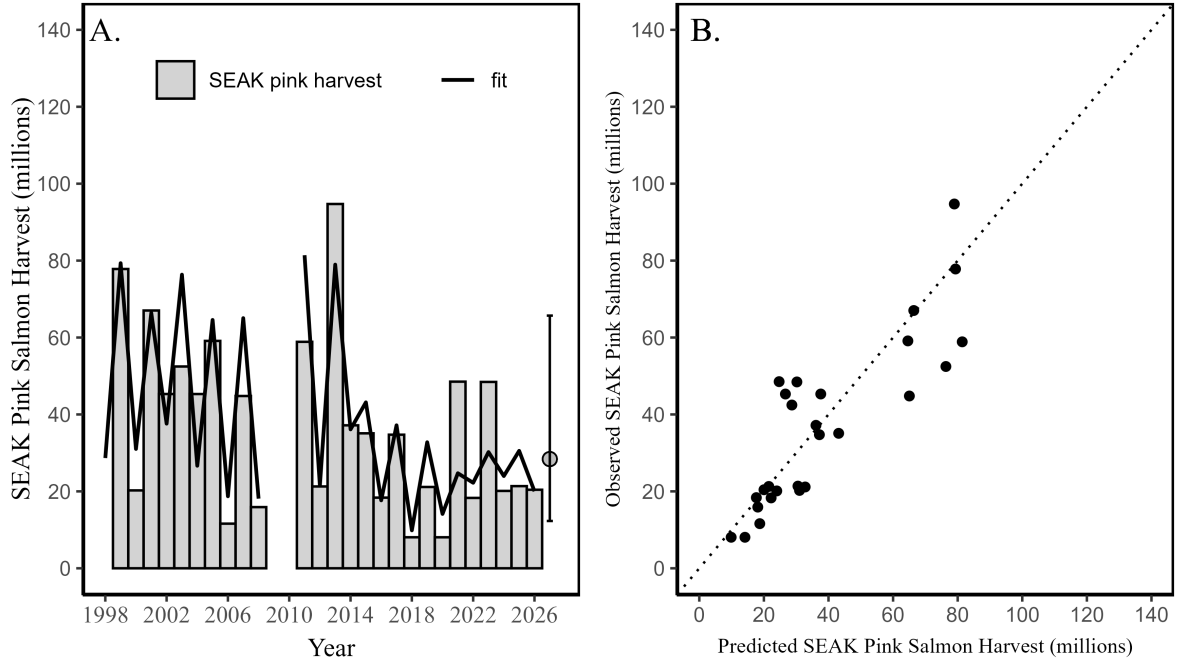


Figure 2: A. SEAK pink salmon harvest (millions) by year with the model fit (line) based upon the best performing model (model m11). The predicted 2027 forecast (28.4 million fish) is symbolized as a grey circle with an 80% prediction interval (12.3 to 65.7 million fish). B. SEAK pink salmon harvest (millions) against the fitted values from model m11 by year. The dotted line is a one to one reference line.

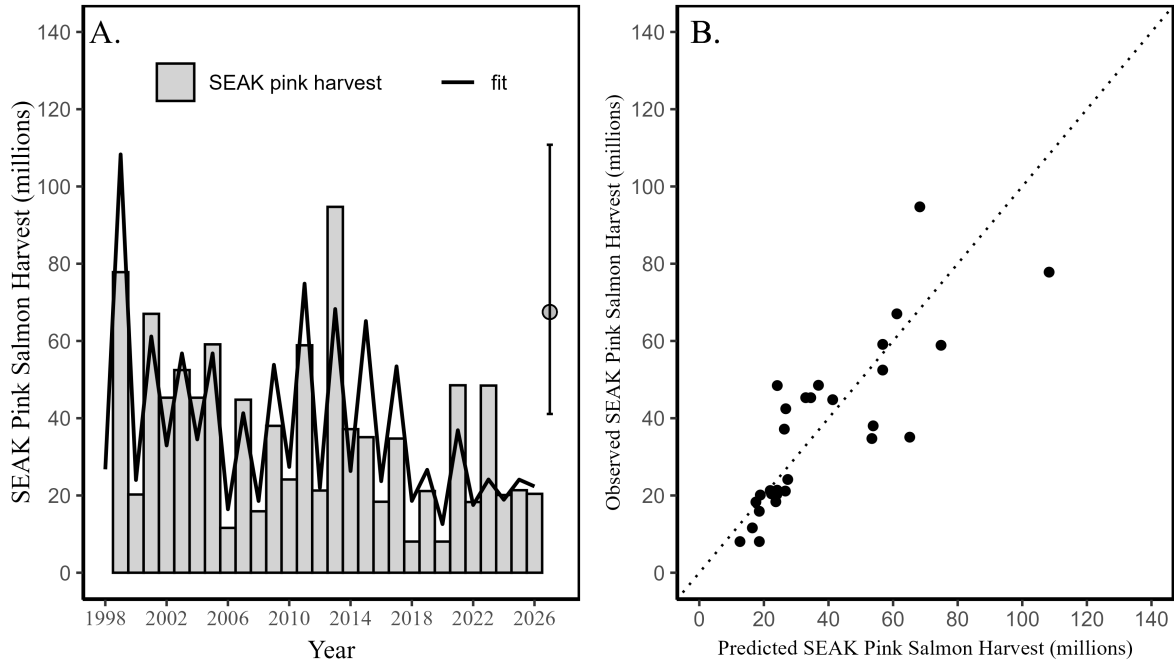


Figure 3: A. SEAK pink salmon harvest (millions) by year with the model fit (line) based upon the best performing model (model m2a). The predicted 2027 forecast (67.5 million fish) is symbolized as a grey circle with an 80% prediction interval (41.1 to 110.8 million fish). B. SEAK pink salmon harvest (millions) against the fitted values from model m2a by year. The dotted line is a one to one reference line.

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Appendix A

Variable definitions

adj_raw_pink_log: First, the raw juvenile pink salmon catch by year, month, haul, and area was adjusted to a 20 minute haul (raw_pink_catch). Then, these values were log-transformed (i.e., $\ln(\text{raw_pink_catch} + 1)$). Next, the maximum value in either June or July, whichever month had the highest in a given year, was used as the adj_raw_pink_log value for the year (i.e., $\text{adj_raw_pink_log} = \text{maximum}(\ln(\text{raw_pink_catch} + 1))$ in either June or July).

vessel-adjusted CPUE: The average $\ln(\text{CPUE}+1)$ for juvenile pink salmon catches in either June or July, whichever month had the highest average in a given year, where effort was a standard trawl haul (20 minutes). The juvenile CPUE data was adjusted using vessel calibration factors to account for differences in fishing power among vessels.

ISTI20_JJ: The average 20-m integrated water column temperature at the eight stations in Icy Strait (Icy Strait and Upper Chatham transects) sampled during the SECM surveys in June and July of each year (in degrees Celsius). In 2025, there was no survey in May. Therefore, this temperature index no longer includes May in the time series temperature index.

Satellite SST variables

Icy_Strait_SST_May: The Icy Strait region encompasses waters of Icy Strait from the east end of Lemesurier Island to a line from Point Couverden south to Point Augusta. This variable is the average SST in May.

Icy_Strait_SST_MJJ: The Icy Strait region encompasses waters of Icy Strait from the east end of Lemesurier Island to a line from Point Couverden south to Point Augusta. This variable is the average SST in May through July.

Icy_Strait_SST_AMJ: The Icy Strait region encompasses waters of Icy Strait from the east end of Lemesurier Island to a line from Point Couverden south to Point Augusta. This variable is the average SST in April through June.

Icy_Strait_SST_AMJJ: The Icy Strait region encompasses waters of Icy Strait from the east end of Lemesurier Island to a line from Point Couverden south to Point Augusta. This variable is the average SST in April through July.

Chatham_SST_May: The Chatham and Icy Straits region encompasses waters of Chatham and Icy Straits east of Lemesurier Island to Point Couverden, and south to the approximate latitude of 56.025 degrees north (roughly Cape Decision off Kuiu Island). This variable is the average SST in May.

Chatham_SST_MJJ: The Chatham and Icy Straits region encompasses waters of Chatham and Icy Straits east of Lemesurier Island to Point Couverden, south to the approximate latitude of 56.025 degrees north (roughly Cape Decision off Kuiu Island). This variable is the average SST in May through July.

Chatham_SST_AMJ: The Chatham and Icy Straits region encompasses waters of Chatham and Icy Straits east of Lemesurier Island to Point Couverden, south to the approximate latitude of 56.025 degrees north (roughly Cape Decision off Kuiu Island). This variable is the average SST in April through June. The last time this variable was incorporated in the forecasting process was the 2023 forecast.

Chatham_SST_AMJJ: The Chatham and Icy Straits region encompasses waters of Chatham and Icy Straits east of Lemesurier Island to Point Couverden, south to the approximate latitude of 56.025 degrees north (roughly Cape Decision off Kuiu Island). This variable is the average SST in April through July.

NSEAK_SST_May: The NSEAK region encompasses northern Southeast Alaska from 59.475 to 56.075 degrees north latitude (approximately Districts 9 through 15, and District 13 inside area only; northern Southeast Inside subregion for Southeast Alaska (NSEI)). This variable is the average SST in May.

NSEAK_SST_MJJ: The NSEAK region encompasses northern Southeast Alaska from 59.475 to 56.075 degrees north latitude (approximately Districts 9 through 15, and District 13 inside area only; northern Southeast Inside subregion for Southeast Alaska (NSEI)). This variable is the average SST in May through July.

NSEAK_SST_AMJ: The NSEAK region encompasses northern Southeast Alaska from 59.475 to 56.075 degrees north latitude (approximately Districts 9 through 15, and District 13 inside area only; northern Southeast Inside subregion for Southeast Alaska (NSEI)). This variable is the average SST in April through June.

NSEAK_SST_AMJJ: The NSEAK region encompasses northern Southeast Alaska from 59.475 to 56.075 degrees north latitude (approximately Districts 9 through 15, and District 13 inside area only; northern Southeast Inside subregion for Southeast Alaska (NSEI)). This variable is the average SST in April through July.

SEAK_SST_May: The SEAK region encompasses Southeast Alaska from 59.475 to 54.725 degrees north latitude. This variable is the average SST in May. The last time this variable was incorporated in the forecasting process was the 2023 forecast.

SEAK_SST_MJJ: The SEAK region encompasses northern Southeast Alaska from 59.475 to 54.725 degrees north latitude. This variable is the average SST in May through July.

SEAK_SST_AMJ: The SEAK region encompasses Southeast Alaska from 59.475 to 54.725 degrees north latitude. This variable is the average SST in April through June.

SEAK_SST_AMJJ: The SEAK region encompasses Southeast Alaska from 59.475 to 54.725 degrees north latitude. This variable is the average SST in April through July.

Appendix B

Table 4: Parameter estimates for the 18 individual models for Models m1-m18.

Model	Term	Estimate	Standard Error	Statistic	p value
m1	(Intercept)	1.306	0.966	1.353	0.191
m1	as.factor(odd_even_factor)odd	0.351	0.231	1.522	0.144
m1	as.factor(vessel)Medeia	2.585	1.668	1.550	0.137
m1	as.factor(vessel)NW Explorer	-0.118	1.063	-0.111	0.912
m1	adj_raw_pink_log	0.381	0.179	2.134	0.045
m1	as.factor(vessel)Medeia:adj_raw_pink_log	-0.601	0.342	-1.760	0.094
m1	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.088	0.175	-0.503	0.621
m2	(Intercept)	4.310	1.735	2.485	0.022
m2	as.factor(odd_even_factor)odd	0.220	0.224	0.982	0.339
m2	as.factor(vessel)Medeia	2.651	1.552	1.708	0.104
m2	as.factor(vessel)NW Explorer	-0.357	0.996	-0.358	0.724
m2	ISTI20_JJ	-0.324	0.160	-2.025	0.057
m2	adj_raw_pink_log	0.438	0.168	2.599	0.018
m2	as.factor(vessel)Medeia:adj_raw_pink_log	-0.610	0.318	-1.919	0.070
m2	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.072	0.163	-0.441	0.664
m3	(Intercept)	3.410	1.224	2.785	0.012
m3	as.factor(odd_even_factor)odd	0.362	0.207	1.751	0.096
m3	as.factor(vessel)Medeia	2.532	1.495	1.694	0.107
m3	as.factor(vessel)NW Explorer	0.057	0.955	0.060	0.953
m3	Chatham_SST_May	-0.309	0.127	-2.430	0.025
m3	adj_raw_pink_log	0.410	0.160	2.554	0.019
m3	as.factor(vessel)Medeia:adj_raw_pink_log	-0.559	0.307	-1.822	0.084
m3	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.098	0.157	-0.622	0.541
m4	(Intercept)	3.944	1.698	2.323	0.031
m4	as.factor(odd_even_factor)odd	0.338	0.218	1.551	0.137
m4	as.factor(vessel)Medeia	2.636	1.576	1.672	0.111
m4	as.factor(vessel)NW Explorer	0.078	1.010	0.078	0.939
m4	Chatham_SST_MJJ	-0.291	0.158	-1.842	0.081
m4	adj_raw_pink_log	0.416	0.170	2.452	0.024
m4	as.factor(vessel)Medeia:adj_raw_pink_log	-0.587	0.323	-1.816	0.085
m4	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.115	0.166	-0.690	0.499
m5	(Intercept)	4.071	1.368	2.976	0.008
m5	as.factor(odd_even_factor)odd	0.346	0.204	1.700	0.105
m5	as.factor(vessel)Medeia	2.403	1.474	1.630	0.119
m5	as.factor(vessel)NW Explorer	0.118	0.943	0.126	0.901
m5	Chatham_SST_AMJ	-0.394	0.152	-2.584	0.018

Model	Term	Estimate	Standard Error	Statistic	p value
m5	adj_raw_pink_log	0.417	0.158	2.639	0.016
m5	as.factor(vessel)Medeia:adj_raw_pink_log	-0.529	0.303	-1.747	0.097
m5	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.108	0.155	-0.700	0.492
m6	(Intercept)	4.245	1.573	2.699	0.014
m6	as.factor(odd_even_factor)odd	0.333	0.210	1.586	0.129
m6	as.factor(vessel)Medeia	2.552	1.520	1.679	0.110
m6	as.factor(vessel)NW Explorer	0.146	0.976	0.149	0.883
m6	Chatham_SST_AMJJ	-0.370	0.164	-2.255	0.036
m6	adj_raw_pink_log	0.428	0.164	2.608	0.017
m6	as.factor(vessel)Medeia:adj_raw_pink_log	-0.561	0.312	-1.798	0.088
m6	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.124	0.160	-0.770	0.451
m7	(Intercept)	3.228	1.193	2.707	0.014
m7	as.factor(odd_even_factor)odd	0.393	0.209	1.884	0.075
m7	as.factor(vessel)Medeia	2.462	1.506	1.635	0.118
m7	as.factor(vessel)NW Explorer	-0.048	0.960	-0.050	0.960
m7	Icy_Strait_SST_May	-0.294	0.125	-2.360	0.029
m7	adj_raw_pink_log	0.401	0.161	2.489	0.022
m7	as.factor(vessel)Medeia:adj_raw_pink_log	-0.548	0.309	-1.772	0.092
m7	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.080	0.158	-0.509	0.617
m8	(Intercept)	3.685	1.577	2.337	0.031
m8	as.factor(odd_even_factor)odd	0.371	0.218	1.702	0.105
m8	as.factor(vessel)Medeia	2.442	1.577	1.549	0.138
m8	as.factor(vessel)NW Explorer	0.043	1.008	0.043	0.966
m8	Icy_Strait_SST_MJJ	-0.254	0.137	-1.849	0.080
m8	adj_raw_pink_log	0.407	0.169	2.405	0.027
m8	as.factor(vessel)Medeia:adj_raw_pink_log	-0.554	0.324	-1.711	0.103
m8	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.110	0.166	-0.661	0.516
m9	(Intercept)	3.877	1.351	2.868	0.010
m9	as.factor(odd_even_factor)odd	0.400	0.207	1.933	0.068
m9	as.factor(vessel)Medeia	2.300	1.493	1.540	0.140
m9	as.factor(vessel)NW Explorer	0.026	0.951	0.027	0.978
m9	Icy_Strait_SST_AMJ	-0.374	0.151	-2.470	0.023
m9	adj_raw_pink_log	0.406	0.160	2.542	0.020
m9	as.factor(vessel)Medeia:adj_raw_pink_log	-0.510	0.307	-1.661	0.113
m9	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.093	0.156	-0.598	0.557
m10	(Intercept)	3.889	1.481	2.626	0.017
m10	as.factor(odd_even_factor)odd	0.371	0.212	1.752	0.096
m10	as.factor(vessel)Medeia	2.434	1.532	1.588	0.129
m10	as.factor(vessel)NW Explorer	0.124	0.982	0.127	0.901
m10	Icy_Strait_SST_AMJJ	-0.326	0.150	-2.177	0.042
m10	adj_raw_pink_log	0.423	0.165	2.563	0.019

Model	Term	Estimate	Standard Error	Statistic	p value
m10	as.factor(vessel)Medeia:adj_raw_pink_log	-0.543	0.315	-1.724	0.101
m10	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.122	0.162	-0.756	0.459
m11	(Intercept)	3.336	1.196	2.789	0.012
m11	as.factor(odd_even_factor)odd	0.386	0.207	1.868	0.077
m11	as.factor(vessel)Medeia	2.476	1.492	1.659	0.113
m11	as.factor(vessel)NW Explorer	0.139	0.956	0.146	0.886
m11	NSEAK_SST_May	-0.297	0.121	-2.453	0.024
m11	adj_raw_pink_log	0.400	0.160	2.500	0.022
m11	as.factor(vessel)Medeia:adj_raw_pink_log	-0.537	0.307	-1.752	0.096
m11	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.106	0.157	-0.675	0.508
m12	(Intercept)	3.947	1.629	2.422	0.026
m12	as.factor(odd_even_factor)odd	0.350	0.216	1.622	0.121
m12	as.factor(vessel)Medeia	2.654	1.563	1.698	0.106
m12	as.factor(vessel)NW Explorer	0.193	1.008	0.191	0.851
m12	NSEAK_SST_MJJ	-0.294	0.151	-1.949	0.066
m12	adj_raw_pink_log	0.418	0.168	2.485	0.022
m12	as.factor(vessel)Medeia:adj_raw_pink_log	-0.574	0.320	-1.790	0.089
m12	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.128	0.165	-0.772	0.450
m13	(Intercept)	3.982	1.343	2.966	0.008
m13	as.factor(odd_even_factor)odd	0.368	0.204	1.805	0.087
m13	as.factor(vessel)Medeia	2.439	1.474	1.655	0.114
m13	as.factor(vessel)NW Explorer	0.245	0.949	0.258	0.799
m13	NSEAK_SST_AMJ	-0.391	0.152	-2.580	0.018
m13	adj_raw_pink_log	0.416	0.158	2.628	0.017
m13	as.factor(vessel)Medeia:adj_raw_pink_log	-0.521	0.303	-1.718	0.102
m13	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.123	0.155	-0.792	0.438
m14	(Intercept)	4.179	1.535	2.722	0.014
m14	as.factor(odd_even_factor)odd	0.352	0.210	1.680	0.109
m14	as.factor(vessel)Medeia	2.594	1.516	1.711	0.103
m14	as.factor(vessel)NW Explorer	0.263	0.981	0.268	0.791
m14	NSEAK_SST_AMJJ	-0.367	0.161	-2.281	0.034
m14	adj_raw_pink_log	0.428	0.164	2.618	0.017
m14	as.factor(vessel)Medeia:adj_raw_pink_log	-0.554	0.311	-1.780	0.091
m14	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.138	0.161	-0.859	0.401
m15	(Intercept)	3.304	1.218	2.713	0.014
m15	as.factor(odd_even_factor)odd	0.364	0.208	1.750	0.096
m15	as.factor(vessel)Medeia	2.690	1.507	1.785	0.090
m15	as.factor(vessel)NW Explorer	0.198	0.969	0.204	0.841
m15	SEAK_SST_May	-0.282	0.120	-2.350	0.030
m15	adj_raw_pink_log	0.416	0.162	2.572	0.019
m15	as.factor(vessel)Medeia:adj_raw_pink_log	-0.578	0.309	-1.873	0.076

Model	Term	Estimate	Standard Error	Statistic	p value
m15	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.117	0.159	-0.740	0.469
m16	(Intercept)	3.976	1.612	2.466	0.023
m16	as.factor(odd_even_factor)odd	0.329	0.215	1.529	0.143
m16	as.factor(vessel)Medeia	2.810	1.560	1.801	0.088
m16	as.factor(vessel)NW Explorer	0.223	1.006	0.222	0.827
m16	SEAK_SST_MJJ	-0.288	0.144	-1.997	0.060
m16	adj_raw_pink_log	0.426	0.168	2.537	0.020
m16	as.factor(vessel)Medeia:adj_raw_pink_log	-0.597	0.319	-1.872	0.077
m16	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.130	0.165	-0.787	0.441
m17	(Intercept)	3.943	1.361	2.897	0.009
m17	as.factor(odd_even_factor)odd	0.348	0.205	1.697	0.106
m17	as.factor(vessel)Medeia	2.594	1.484	1.747	0.097
m17	as.factor(vessel)NW Explorer	0.311	0.962	0.323	0.750
m17	SEAK_SST_AMJ	-0.369	0.148	-2.499	0.022
m17	adj_raw_pink_log	0.433	0.160	2.703	0.014
m17	as.factor(vessel)Medeia:adj_raw_pink_log	-0.547	0.305	-1.794	0.089
m17	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.136	0.157	-0.867	0.396
m18	(Intercept)	4.211	1.540	2.735	0.013
m18	as.factor(odd_even_factor)odd	0.335	0.209	1.602	0.126
m18	as.factor(vessel)Medeia	2.736	1.516	1.806	0.087
m18	as.factor(vessel)NW Explorer	0.306	0.983	0.311	0.759
m18	SEAK_SST_AMJJ	-0.356	0.155	-2.295	0.033
m18	adj_raw_pink_log	0.437	0.164	2.668	0.015
m18	as.factor(vessel)Medeia:adj_raw_pink_log	-0.574	0.310	-1.849	0.080
m18	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.142	0.161	-0.884	0.388

Table 5: Parameter estimates for the 18 individual models for Models m1a-m18a.

Model	Term	Estimate	Standard Error	Statistic	p value
m1a	(Intercept)	2.496	0.163	15.298	0.000
m1a	CPUE	0.304	0.072	4.201	0.000
m1a	as.factor(odd_even_factor)odd	0.400	0.175	2.289	0.030
m2a	(Intercept)	6.028	1.353	4.456	0.000
m2a	CPUE	0.396	0.074	5.343	0.000
m2a	as.factor(odd_even_factor)odd	0.196	0.176	1.112	0.277
m2a	ISTI20_JJ	-0.368	0.140	-2.627	0.015
m3a	(Intercept)	5.528	0.693	7.973	0.000
m3a	CPUE	0.363	0.057	6.400	0.000
m3a	as.factor(odd_even_factor)odd	0.357	0.134	2.670	0.013
m3a	Chatham_SST_May	-0.417	0.094	-4.445	0.000
m4a	(Intercept)	5.832	1.221	4.776	0.000
m4a	CPUE	0.334	0.066	5.099	0.000
m4a	as.factor(odd_even_factor)odd	0.346	0.157	2.199	0.037
m4a	Chatham_SST_MJJ	-0.345	0.125	-2.751	0.011
m5a	(Intercept)	6.125	0.852	7.185	0.000
m5a	CPUE	0.360	0.057	6.269	0.000
m5a	as.factor(odd_even_factor)odd	0.332	0.136	2.442	0.022
m5a	Chatham_SST_AMJ	-0.486	0.113	-4.304	0.000
m6a	(Intercept)	6.217	1.089	5.709	0.000
m6a	CPUE	0.345	0.062	5.577	0.000
m6a	as.factor(odd_even_factor)odd	0.335	0.148	2.260	0.033
m6a	Chatham_SST_AMJJ	-0.434	0.126	-3.445	0.002
m7a	(Intercept)	5.221	0.647	8.067	0.000
m7a	CPUE	0.370	0.058	6.373	0.000
m7a	as.factor(odd_even_factor)odd	0.384	0.135	2.837	0.009
m7a	Icy_Strait_SST_May	-0.399	0.093	-4.293	0.000
m8a	(Intercept)	5.561	1.105	5.032	0.000
m8a	CPUE	0.341	0.066	5.193	0.000
m8a	as.factor(odd_even_factor)odd	0.362	0.156	2.320	0.029
m8a	Icy_Strait_SST_MJJ	-0.313	0.112	-2.798	0.010
m9a	(Intercept)	5.874	0.822	7.144	0.000
m9a	CPUE	0.361	0.058	6.186	0.000
m9a	as.factor(odd_even_factor)odd	0.388	0.137	2.831	0.009
m9a	Icy_Strait_SST_AMJ	-0.471	0.113	-4.159	0.000
m10a	(Intercept)	5.876	1.012	5.806	0.000
m10a	CPUE	0.350	0.063	5.588	0.000
m10a	as.factor(odd_even_factor)odd	0.365	0.148	2.461	0.021
m10a	Icy_Strait_SST_AMJJ	-0.398	0.118	-3.371	0.002

Model	Term	Estimate	Standard Error	Statistic	p value
m11a	(Intercept)	5.265	0.648	8.126	0.000
m11a	CPUE	0.334	0.056	5.960	0.000
m11a	as.factor(odd_even_factor)odd	0.393	0.134	2.924	0.007
m11a	NSEAK_SST_May	-0.378	0.087	-4.357	0.000
m12a	(Intercept)	5.629	1.118	5.034	0.000
m12a	CPUE	0.311	0.064	4.836	0.000
m12a	as.factor(odd_even_factor)odd	0.382	0.155	2.459	0.021
m12a	NSEAK_SST_MJJ	-0.318	0.112	-2.826	0.009
m13a	(Intercept)	5.857	0.819	7.153	0.000
m13a	CPUE	0.328	0.057	5.752	0.000
m13a	as.factor(odd_even_factor)odd	0.384	0.137	2.801	0.010
m13a	NSEAK_SST_AMJ	-0.450	0.108	-4.156	0.000
m14a	(Intercept)	5.924	1.029	5.755	0.000
m14a	CPUE	0.318	0.061	5.188	0.000
m14a	as.factor(odd_even_factor)odd	0.380	0.148	2.566	0.017
m14a	NSEAK_SST_AMJJ	-0.397	0.118	-3.361	0.002
m15a	(Intercept)	5.266	0.701	7.512	0.000
m15a	CPUE	0.329	0.058	5.699	0.000
m15a	as.factor(odd_even_factor)odd	0.398	0.139	2.862	0.008
m15a	SEAK_SST_May	-0.349	0.087	-4.021	0.000
m16a	(Intercept)	5.524	1.117	4.948	0.000
m16a	CPUE	0.300	0.065	4.640	0.000
m16a	as.factor(odd_even_factor)odd	0.392	0.156	2.505	0.019
m16a	SEAK_SST_MJJ	-0.290	0.106	-2.736	0.011
m17a	(Intercept)	5.787	0.866	6.686	0.000
m17a	CPUE	0.322	0.059	5.489	0.000
m17a	as.factor(odd_even_factor)odd	0.389	0.141	2.755	0.011
m17a	SEAK_SST_AMJ	-0.408	0.106	-3.847	0.001
m18a	(Intercept)	5.801	1.044	5.558	0.000
m18a	CPUE	0.307	0.062	4.941	0.000
m18a	as.factor(odd_even_factor)odd	0.393	0.150	2.619	0.015
m18a	SEAK_SST_AMJJ	-0.358	0.112	-3.195	0.004

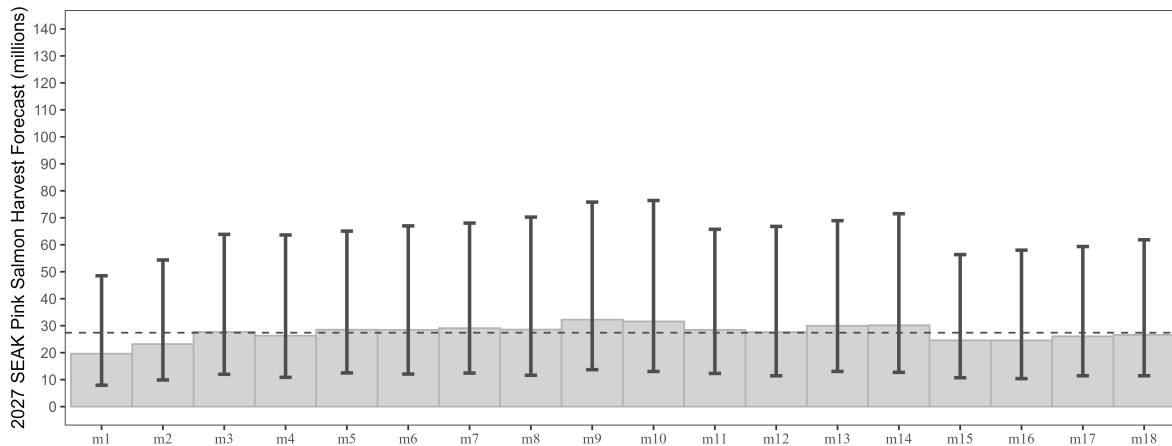


Figure B1: Bias-corrected forecasts (grey bars) for the 18 regression models (Models m1-m18) with 80% prediction intervals (vertical grey lines). A horizontal dotted line at the average model results is placed on the figure for reference only. The 2027-forecast using the model mxx has a point estimate of xx million fish (80% prediction interval: xx to xx million fish).

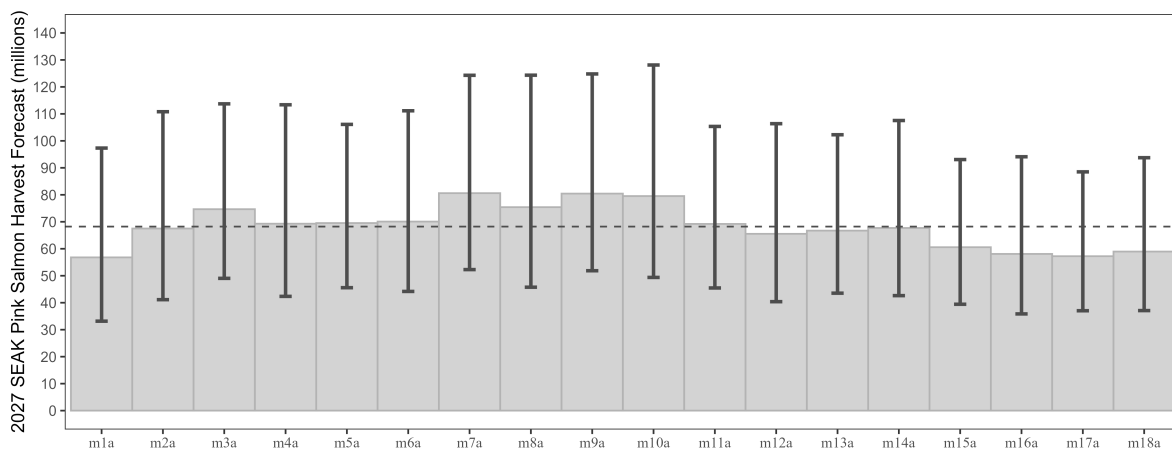


Figure B2: Bias-corrected forecasts (grey bars) for the 18 regression models (Models m1a-m18a) with 80% prediction intervals (vertical grey lines). A horizontal dotted line at the average model results is placed on the figure for reference only. The 2027-forecast using the model mxx has a point estimate of xx million fish (80% prediction interval: xx to xx million fish).